INDEX · 10 LESSONS · 144 FILES

Python Practice Programs

Demo.py → L10 / guess_number-simp.py

WHAT'S IN THE ZIP

144 files across 10 lessons

131

Python files

129 in lessons, plus 2 loose scripts

10

Lessons

From Hello World to object-oriented code

13

Other files

Text data files for file handling (L7)

How the files are named

`L9/Q12.py = Lesson 9, Question 12 · Qh = homework`

Lesson map

Lesson	Topic	Files
L1	Python basics: output, variables, operators	16
L2	Strings and conditions	16
L3	Lists and tuples	9
L4	Dictionaries and sets	14
L5	Loops: while, for, range	18
L6	Functions and recursion	13
L7	File handling	14
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L9	Object-oriented Python in depth	18
L10	Mini projects	4

LESSON 1 · 16 FILES · PART 1 OF 2

Printing, types and basic operators

File	What it does
Hello.py	Hello World
Q1.py	print() with commas and +
Q2.py	Variables, data types, quote styles
Q3.py	Printing types with type()

File	What it does
Q4.py	Sums of predefined values
Q5.py	Single and multi-line comments
Q6.py	Arithmetic operators (/, //, **, %)
Q7.py	Relational operators

LESSON 1 · 16 FILES · PART 2 OF 2

Operators, casting, input and homework

File	What it does
Q8.py	Assignment operators (+=, *=, **=)
Q9.py	Logical operators: not, or, and
Q10.py	Type conversion and type casting
Q11.py	input() as int, float, string

File	What it does
Qh1.py	Sum of two numbers
Qh2.py	Area of a square
Qh3.py	Average of two floats
Qh4.py	Is first number >= second?

LESSON 2 · 16 FILES · PART 1 OF 2

Strings

File	What it does
Q1.py	String quote styles and \n
Q2.py	Concatenation
Q3.py	Length with len()
Q4.py	Indexing (strings are immutable)

File	What it does
Q5.py	Slicing [start:end]
Q6.py	Negative indexes
Q7.py	String methods: replace, find, count

LESSON 2 · 16 FILES · PART 2 OF 2

Conditions and homework

File	What it does
Q8.py	if / elif / else age groups
Q9.py	Grade calculator
Q10.py	Nested if: roller coaster check
Qh1.py	Length of an input string
Qh2.py	Count of $ in a string

File	What it does
Qh3.py	Even or odd
Qh4.py	Greatest of three numbers
Qh5.py	Multiple of 7 check
Qh6.py	Greatest of four (empty stub)

LESSON 3 · 9 FILES

Lists and tuples

File	What it does
Q1.py	Lists: indexing, types, changing items
Q2.py	List slicing and steps
Q3.py	append, sort, reverse, insert, pop
Q4.py	Tuples, empty and single-item
Q5.py	Tuple slicing

File	What it does
Q6.py	Tuple count() and index()
Qh1.py	Read 3 favourite movies into a list
Qh2.py	Is a list a palindrome?
Qh3.py	Count grade A in a tuple

LESSON 4 · 14 FILES · PART 1 OF 2

Dictionaries

File	What it does
Q1.py	Dictionary with a student record
Q2.py	Update, add keys, append to a list value
Q3.py	Empty dictionary
Q4.py	Nested dictionary of employees

File	What it does
Q5i.py	keys, values, items, get, update, pop
Q5ii.py	Same methods, inline comments
Q6.py	get() returns None, not an error

LESSON 4 · 14 FILES · PART 2 OF 2

Sets and homework

File	What it does
Q7.py	Sets: unique, unordered
Q8.py	Empty set vs empty dict
Q9.py	add, remove, discard, union, intersection
Qh1.py	Store word meanings in a dict

File	What it does
Qh2.py	Classrooms needed: unique subjects
Qh3.py	Marks of 3 subjects into a dict
Qh4.py	Store 9 and 9.0 as separate set items

LESSON 5 · 18 FILES · PART 1 OF 2

while, for, range and early exercises

File	What it does
Q1.py	while loop
Q2.py	break and continue
Q3.py	for loop over a list and range
Q4.py	range(start, stop, step)
Q5.py	pass placeholder

File	What it does
Qh1.py	Print 1 to 100 (while)
Qh2.py	Print 100 to 1 (while)
Qh3.py	Multiplication table (while)
Qh4.py	Print a list with while
Qh5.py	Search a tuple, while-else

LESSON 5 · 18 FILES · PART 2 OF 2

Patterns, for-loops and tables

File	What it does
Qh6.py	Right-angled triangle of stars
Qh7.py	Print a list with for
Qh8.py	Search a tuple, for-else
Qh9.py	Print 1 to 100 (for, range)

File	What it does
Qh10.py	Print 100 to 1 (for, range)
Qh11.py	Multiplication table (for, range)
Qh12.py	Sum of 1 to n (while-else)
Qh13.py	Factorial of n (for)

LESSON 6 · 13 FILES · PART 1 OF 2

Functions: basics and recursion

File	What it does
Q1.py	Define and call a function
Q2.py	Parameters, arguments, return
Q3.py	Recursion: count down and up
Q4.py	Factorial using recursion

File	What it does
Qh1.py	Average of three numbers
Qh2.py	Print length of a list
Qh3.py	Print list items on one line

LESSON 6 · 13 FILES · PART 2 OF 2

Functions: exercises

File	What it does
Qh4.py	Factorial, loop and recursive
Qh5.py	USD and INR converter
Qh6.py	Return ODD or EVEN

File	What it does
Qh7.py	Recursive sum of first n numbers
Qh8.py	Print a list recursively
Qh9.py	Same, printed in order

Opening, reading and writing

File	What it does
Q1.py	Open, read, close
Q2.py	read, seek, readline
Q3.py	Write mode and writelines
Q4.py	Append mode
Q5.py	w+ : write, then read

File	What it does
Q6.py	r+ : read and write
Q7.py	a+ with tell()
Q8.py	with open(...) as file
Q9.py	Delete a file with os.remove

Homework

File	What it does
`Qh1.py`	Write four lines to a file
`Qh2.py`	Replace Java with Python
`Qh3.py`	Is "learning" in the file?

File	What it does
`Qh4.py`	Line number of first "learning"
`Qh5.py`	Count even numbers in a file

Each program reads or writes a matching .txt file of the same name (Q3.txt, Qh5.txt and so on).

LESSON 8 · 7 FILES

Classes and objects

File	What it does
Q1.py	First class and object
Q2.py	Class vs instance attributes
Q3.py	__init__ constructor
Q4.py	Methods and self

File	What it does
Q5.py	@staticmethod
Qh1.py	Student class with average marks
Qh2.py	Account: debit, credit, balance

LESSON 9 · 18 FILES · PART 1 OF 2

Deleting, privacy and inheritance

File	What it does
Q1.py	del on objects and attributes
Q2.py	Re-assign after del
Q3.py	Private attribute (__)
Q4.py	Private method (__)
Q5.py	Inheritance

File	What it does
Q6.py	Single inheritance
Q7.py	Multi-level inheritance
Q8.py	Multiple inheritance
Q9.py	super() in a subclass

LESSON 9 · 18 FILES · PART 2 OF 2

Decorators, polymorphism and homework

File	What it does
Q10.py	@classmethod
Q11.py	@property for a computed value
Q12.py	Getter and setter: bank balance
Q13.py	Getter and setter: temperature
Q14.py	Setter with mark validation

File	What it does
Q15.py	Operator overloading: Complex numbers
Qh1.py	Circle: area and perimeter
Qh2.py	Engineer inherits Employee
Qh3.py	Order compare with __gt__

LESSON 10 · 4 FILES

Mini projects

File	What it does
`Random_pwd_gen.py`	Password generator using secrets
`Random_pwd_gen_simp.py`	Password generator, simpler version

File	What it does
`guess_number.py`	Guessing game as a class
`guess_number-simp.py`	Guessing game, simpler version

Each game and generator comes in two versions: a full one and a "-simp" one.

OUTSIDE THE LESSONS · 15 FILES

Extra programs and data files

File	What it is
`Demo.py`	Empty file
`F_Strings.py`	f-string examples with variables and a price total (15 lines)
`L7 / Q1 to Q8.txt`	Data files for Lesson 7 (8 files; Q2.txt is empty)
`L7 / Qh1 to Qh5.txt`	Homework data files for Lesson 7 (5 files)